Module - ST4003CEM Object Oriented Programming
Module leader- Prabisha Khadka’
Submitted by- Avigya Sharma

Q1) Write a program to check whether a person can cast a vote or not. The condition is you must be over 18 years old to vote.

Ans: Source code

Task1.java

```java
import java.util.Scanner;
public class Task1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();
        if (age >= 18) {
            System.out.println("You can vote.");
        } else {
            System.out.println("You cannot vote.");
        }
    }
}
```

Output

```
Enter your age: 12
You cannot vote.

=== Code Execution Successful ===
```

Q2)Write a program to calculate Simple Interest. Formula Simple Interest = (PrincipleAmount*Time*Rate/100);

Ans: Source code:

Task2.java

```java
import java.util.Scanner;

public class Task2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Principal: ");
        double p = sc.nextDouble();

        System.out.print("Enter Time: ");
        double t = sc.nextDouble();

        System.out.print("Enter Rate: ");
        double r = sc.nextDouble();

        double si = (p*t*r)/100;
        System.out.println("Simple Interest="+si);
    }
}
```

Output

```
Output
Enter Principal: 2
Enter Time: 1
Enter Rate: 2
Simple Interest=0.04

=== Code Execution Successful ===
```

Q3)Write a program to calculate, area of the triangle, and the Volume of the Cube and Cuboid. **Formula: Volume of Cuboid = length*width*height; and Volume of Cube: side*side*side;**

Ans: Source code

Task.java

```java
import java.util.Scanner;

public class Task {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter breadth: ");
        float breadth = sc.nextFloat();
        System.out.print("Enter height: ");
        float height = sc.nextFloat();
        float areaTriangle = 0.5f *  breadth * height;

        System.out.println("Area of Triangle = " + areaTriangle);
        System.out.print("Enter side of cube: ");
        float side = sc.nextFloat();
        float volumeCube = side * side * side;
        System.out.println("Volume of Cube = " + volumeCube);

        System.out.print("Enter length, width, height: ");
        float l = sc.nextFloat();
        float w = sc.nextFloat();
        float h = sc.nextFloat();
        float volumeCuboid = l * w * h;
        System.out.println("Volume of Cuboid = " + volumeCuboid);
    }
}
```

Output

```
Output
Enter breadth: 2
Enter height: 2
Area of Triangle = 2.0
Enter side of cube: 2
Volume of Cube = 8.0
Enter length, width, height: 3
2
3
Volume of Cuboid = 18.0

=== Code Execution Successful ===
```

Q4)Write the ternary operator for question no. 1

Ans: Source code

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        votingRule();
    }
    static void votingRule() {
    System.out.println("Enter your age: ");
    int age = sc.nextInt();

    String check = (age >= 18) ? "eligible to vote" : "not eligible to vote";

    System.out.println("You are " + check);
}
}
```

Output:

Output

```
Enter your age:
18
You are eligible to vote

=== Code Execution Successful ===
```

Q5)Write a program to take two integer inputs from the user and print the sum and product of them.

Ans: Source code

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        sumProduct();
    }
    static void sumProduct() {
        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();
        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();
        int sum = num1 + num2;
        System.out.println("Sum of numbers " + num1 + " and " + num2 + " is: " + sum);
        int product = num1 * num2;
        System.out.println("Product of numbers " + num1 + " and " + num2 + " is: " + product);
    }
}
```

Output

Output

```
Enter first number: 2
Enter second number: 3
Sum of numbers 2 and 3 is: 5
Product of numbers 2 and 3 is: 6

=== Code Execution Successful ===
```

Q6)Take two integer inputs from the user. First, calculate the sum of two, then the product of two. Finally, calculate the division of the thus obtained sum and product and print the result.
Ans: Source code

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        divisionTwo();
    }
    static void divisionTwo() {
        System.out.print("Enter first number: ");
        float num1 = sc.nextFloat();

        System.out.print("Enter second number: ");
        float num2 = sc.nextFloat();

        float sum = num1 + num2;
        float product = num1 * num2;
        float division = product / sum;

        System.out.println("Sum = " + sum);
        System.out.println("Product = " + product);
        System.out.println("Division (product/sum) = " + division
            );
    }
}
```

Output:

Output

```
Enter first number: 2
Enter second number: 3
Sum = 5.0
Product = 6.0
Division (product/sum) = 1.2

=== Code Execution Successful ===
```

Q7)Take the name, roll number, and field of interest from the user and print in the format below: Hey, my name is XYZ and my roll number is XYZ. My field of interest are xyz.
Ans: Source code

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        format();
    }
    static void format() {
        System.out.print("Enter your name: ");
        String name = sc.nextLine();
        System.out.print("Enter your roll number: ");
        int rollNumber = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter your field of interest: ");
        String interest = sc.nextLine();
        System.out.println("Hey, my name is " + name +
                " and my roll number is " + rollNumber +
                ". My field of interest is " + interest + ".");
    }
}
```

Output:

Output

```
Enter your name: Avigya
Enter your roll number: 7
Enter your field of interest: IT
Hey, my name is Avigya and my roll number is 7. My field of interest is
    IT.

=== Code Execution Successful ===
```

Q8)Take the side of a square from the user and print the area and perimeter of it. Also, calculate Simple Interest, Area of triangle and Volume of Cube and Cuboid. Take the attributes as user input.

Ans: Source code

AllInOne.java

```java
import java.util.Scanner;

public class AllInOne {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Side of square: ");
        double s = sc.nextDouble();
        System.out.println("Area = " + (s * s));
        System.out.println("Perimeter = " + (4 * s));
        System.out.print("\nPrincipal, Rate, Time: ");
        double p = sc.nextDouble();
        double r = sc.nextDouble();
        double t = sc.nextDouble();
        System.out.println("SI = " + (p * r * t / 100));
        System.out.print("\nBase, Height: ");
        double b = sc.nextDouble();
        double h = sc.nextDouble();
        System.out.println("Area = " + (0.5 * b * h));
        System.out.print("\nCube side: ");
        double c = sc.nextDouble();
        System.out.println("Volume = " + (c * c * c));
        System.out.print("\nLength, Width, Height: ");
        double l = sc.nextDouble();
        double w = sc.nextDouble();
        double ht = sc.nextDouble();
        System.out.println("Volume = " + (l * w * ht));
        sc.close();
    }
```

Output:

```
Side of square: 4
Area = 16.0
Perimeter = 16.0

Principal, Rate, Time: 2
3
4
SI = 0.24

Base, Height: 1
2
Area = 1.0

Cube side: 3
Volume = 27.0

Length, Width, Height: 1

3
3
Volume = 9.0

=== Code Execution Successful ===
```

Q9)Ask the user to give two double inputs for length and breadth of a rectangle and print area type casted to int.

Ans: Source code

Task.java

```java
import java.util.Scanner;
public class Task {
    static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        areaRectangle();
    }
    static void areaRectangle() {
        System.out.println("enter length of rectangle: ");
        double length = sc.nextDouble();
        System.out.println("enter breadth of rectangle: ");
        double breadth = sc.nextDouble();
        int area = (int) Math.round(length * breadth);
        System.out.println("Area of rectangle with length " +
            length + " and breadth " + breadth + " is: " + area);
    }
}
```

Output:

Output

```
enter length of rectangle:

2
enter breadth of rectangle:
2
Area of rectangle with length 2.0 and breadth 2.0 is: 4

=== Code Execution Successful ===
```

Q10)Write a program to calculate the total marks of four subjects of a student and the total percentage secured. And use the following conditions to generate the final result;
a. If equal to or more than 70 -> First Class

b. If more than 59 -> Upper Second Class

c. If more than 49 -> Second class

d. If more than 39 -> Third class and if below than 40 the result is fail.

Ans: Source code

Week2Task.java

```java
public class Week2Task {
    Scanner sc = new Scanner(System.in);
    void totalMarks() {
        System.out.println("Enter marks in Mathematics: ");
        Float mathematics = sc.nextFloat();
        System.out.println("Enter marks in Business: ");
        Float business = sc.nextFloat();
        System.out.println("Enter marks in Software: ");
        Float software = sc.nextFloat();
        System.out.println("Enter marks in Programming: ");
        Float programming = sc.nextFloat();
        Float total=mathematics + business + software+programming;
        Float percentage = (total / 4);
        String result;
        result = ((percentage >= 70) ? "First Class" :
                  (percentage >  59) ? "Upper Second Class" :
                  (percentage >  49) ? "Second Class" :
                  (percentage >  39) ? "Third Class" : "Fail");
        System.out.println("Total: "      + total);
        System.out.println("Percentage: " + percentage + "%");
        System.out.println("Result: "     + result);
    }
    public static void main(String[] args) {
        Week2Task check = new Week2Task();
        check.totalMarks();
        check.sc.close();
    }
}
```

Output:

Output

```
Enter marks in Mathematics:
70
Enter marks in Business:
90
Enter marks in Software:
75
Enter marks in Programming:
89
Total: 324.0
Percentage: 81.0%
Result: First Class

=== Code Execution Successful ===
```